

STT 465

Bayesian Data Analysis

Lecture 01: Introduction to Bayesian Thinking

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Welcome to STT 465

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- **Class:** Monday–Wednesday–Friday, 10:20–11:10 AM
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What Is This Course About?

- Bayesian statistics provides a framework for **learning from data under uncertainty**.
- We combine information we already have with new data.
- We quantify uncertainty about unknown quantities.
- We use the resulting uncertainty to make predictions and decisions.

Prior information + Data $\longrightarrow$ Updated beliefs

What Will We Learn?

- Bayesian thinking and probability
- Prior, likelihood, and posterior distributions
- Bayesian estimation and credible intervals
- Bayesian hypothesis testing and Bayes factors
- Bayesian prediction and decision making
- Monte Carlo and MCMC
- Bayesian regression and hierarchical models
- Bayesian data analysis using R

Where Will We Use Bayesian Statistics?

Bayesian methods are used in many fields:

- Medicine and public health
- Business and marketing
- Engineering
- Environmental science
- Sports analytics
- Machine learning and artificial intelligence

The mathematics is general; the applications are everywhere.

How Will You Be Evaluated?

Component	Weight
Homework	25%
Midterm 1	30%
Midterm 2	30%
Final project	15%

- 5 homework assignments 5% each. Late submissions will get a deduction of 15 % per late day.
- Midterm 1: October 23
- Midterm 2: December 4
- Final project: December 11

The Final Project

The final project should start with an interesting problem.

A good project should connect

Problem $\longrightarrow$ Statistical Model $\longrightarrow$ Meaningful Conclusion

Possible goals include:

- analysis of a real dataset,
- investigation of a Bayesian method,
- scientifically meaningful application,
- evaluation of a statistical technique.

Statistics Is About Unknowns

Suppose

θ = an unknown quantity.

Examples:

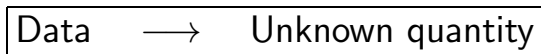
- probability of disease,
- treatment effect,
- probability that a team wins,
- probability that a customer purchases,
- future temperature.

We collect data y and want to learn about θ .

The Fundamental Statistical Problem

We observe data.

We care about something unknown.



Statistical inference is the process of learning about the unknown quantity from the observed data.

But We Rarely Start from Zero

Suppose we want to predict whether a basketball team will win.
Before the next game, we may already know:

- previous wins and losses,
- player performance,
- injuries,
- home-court advantage,
- opponent strength.

Then new information arrives.

Should we ignore everything we knew before?

The Bayesian Idea

Bayesian inference formalizes a natural learning process:

What did I believe before?



What new evidence did I observe?



What should I believe now?

You Already Think Like a Bayesian

Suppose your friend is usually on time.

Today your friend is late.

Do you immediately conclude that your friend is always unreliable?

Probably not.

You combine

past experience + new information.

This is Bayesian thinking.

The Basic Bayesian Framework

$$\text{Prior} + \text{Data} \longrightarrow \text{Posterior}$$

- **Prior:** information before observing the current data
- **Data:** the information we observe
- **Posterior:** updated information after seeing the data

This is the central idea of the course.

A Mysterious Coin

Suppose we have a coin.

Let

$$\theta = P(\text{Heads}).$$

Before flipping the coin, what values of θ seem plausible?

Maybe

$$\theta \approx 0.5$$

is reasonable.

Now suppose we observe

$$H, H, H, H.$$

Should our belief about θ change?

Learning from the Coin

After observing

$H, H, H, H,$

larger values of θ become more plausible.

If instead we observe

$T, T, T, T,$

smaller values of θ become more plausible.

The important idea is:

Data change our uncertainty about θ .
--

What Happens with More Data?

Suppose we observe

80 heads out of 100 flips.

Even if we initially believed

$$\theta \approx 0.5,$$

the data provide strong evidence for larger values of θ .

More data usually means more information.

The Mathematical Idea

The central Bayesian relationship is

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

This connects

- what we believed before,
- what we observed,
- what we believe after seeing the evidence.

In Bayesian statistics,

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior.}$$

Three Important Quantities

For parameter θ and data y :

$$p(\theta)$$

is the **prior distribution**.

$$p(y \mid \theta)$$

is the **likelihood**.

$$p(\theta \mid y)$$

is the **posterior distribution**.

$$p(\theta \mid y) \propto p(y \mid \theta)p(\theta)$$

Prior

The prior describes our uncertainty before seeing the current data.

Prior information may come from:

- previous studies,
- historical data,
- scientific knowledge,
- expert knowledge,
- reasonable assumptions.

A prior is a model for uncertainty before the data.

Likelihood

The likelihood asks:

How compatible is the observed data with different values of θ ?

For example, if

$$\theta = P(\text{Heads}),$$

then observing many heads makes larger values of θ more plausible.

The likelihood is therefore the part of the model where the observed data enter directly.

Posterior

The posterior describes our uncertainty after seeing the data.

$$\text{Posterior} = \text{Updated knowledge}$$

The posterior combines

prior information + evidence from data.

The posterior is the main object of Bayesian inference.

A Surprising Example

Suppose a disease is rare:

$$P(D) = 0.01.$$

A medical test satisfies

$$P(+ \mid D) = 0.99$$

and

$$P(+ \mid D^c) = 0.05.$$

You receive a positive test.

What is

$$P(D \mid +)?$$

Your First Guess?

The test is 99% sensitive.

Maybe the probability of disease after a positive test is

99%.

But is that correct?

Think about the population first.

Think About 10,000 People

Suppose we test 10,000 people.

Because

$$P(D) = 0.01,$$

about

$$100$$

people have the disease.

Among them,

$$0.99(100) = 99$$

test positive.

False Positives Matter

There are

$$10,000 - 100 = 9,900$$

people without the disease.

Among them,

$$0.05(9,900) = 495$$

also test positive.

So the number of positive tests is

$$99 + 495 = 594.$$

The Answer

Among the 594 positive tests, only 99 are from people with the disease.
Therefore,

$$P(D \mid +) = \frac{99}{594} \approx 0.167.$$

About 17%, not 99%.

What Did Bayes' Theorem Do?

Bayes' theorem combined two pieces of information:

Test accuracy

and

Disease prevalence.

In Bayesian language:

Prior information + Observed evidence $\rightarrow$ Posterior probability

This is why conditional probability is central to Bayesian statistics.

Cookies and Candy

Imagine a box containing two types of cookies:

- candy cookies,
- non-candy cookies.

You take one cookie, but you do not know which type it is.

You take one bite.

The bite contains **no candy**.

What should you believe about the cookie?

What Information Matters?

To answer the question, we need to know:

- How common are candy cookies?
- How often does a candy cookie produce a no-candy bite?
- How often does a non-candy cookie produce a no-candy bite?

The answer depends on both the prior composition and the evidence.

Why This Example Matters

The cookie example illustrates a general Bayesian idea:

$$P(\text{cookie type} \mid \text{bite})$$

is not determined by the bite alone.

It also depends on

$$P(\text{cookie type}).$$

Context changes the interpretation of evidence.

Why Do We Need Probability?

Bayesian statistics needs a language for uncertainty.

We need to answer questions such as:

$$P(A)$$

and

$$P(A \mid B).$$

We also need to understand

$$P(A \cap B),$$

$$P(A \cup B),$$

and independence.

These ideas come from probability theory.

Conditional Probability Is the Key

The expression

$$P(A \mid B)$$

means

the probability of A after learning that B occurred.

For Bayesian statistics:

A = unknown quantity

and

B = observed data.

So Bayesian inference is naturally built around conditional probability.

Do Not Confuse These Two

$$P(A \mid B)$$

and

$$P(B \mid A)$$

are generally different.

For example:

$$P(\text{disease} \mid \text{positive test})$$

is not the same as

$$P(\text{positive test} \mid \text{disease}).$$

Bayes' theorem connects them.

Thomas Bayes

- Thomas Bayes lived from 1701 to 1761.
- He was an English minister and mathematician.
- He studied inverse probability.
- His work was published after his death.

His central idea was essentially:

Use observed evidence to learn about an unknown cause.

Pierre-Simon Laplace

- Laplace lived from 1749 to 1827.
- He independently developed and expanded Bayesian ideas.
- He applied probability to scientific problems.
- His work helped establish probability as a tool for inference.

Bayesian inference developed into a general statistical framework.

Why Was Bayesian Statistics Difficult?

The idea is simple:

$$p(\theta \mid y) \propto p(y \mid \theta)p(\theta).$$

But realistic models can involve difficult integrals and high-dimensional calculations.

Before modern computers:

The theory was often easier than the computation.

The Computational Revolution

Modern Bayesian statistics relies heavily on computation.

Important tools include:

- Monte Carlo methods
- Markov chain Monte Carlo
- Gibbs sampling
- Metropolis–Hastings
- modern statistical software

These methods make complex Bayesian models practical.

Bayesian Statistics Today

Bayesian methods are now used for:

- prediction,
- classification,
- regression,
- causal inference,
- hierarchical modeling,
- machine learning,
- decision making.

Bayesian statistics connects probability, statistics, computation, and data science.

Bayesian Thinking in Sports

Suppose a basketball player makes

$$8/10$$

three-point shots.

Is the player's true shooting probability exactly

$$0.80?$$

Not necessarily.

We also know something about typical shooting percentages.

Bayesian analysis combines

previous information + new performance.

Bayesian Thinking in Medicine

Suppose a medical test is positive.

We want

$$P(\text{disease} \mid \text{positive test}).$$

This depends on:

- disease prevalence,
- sensitivity,
- specificity.

Bayesian statistics gives a natural framework for combining them.

Bayesian Thinking in AI

Modern prediction systems often need to answer:

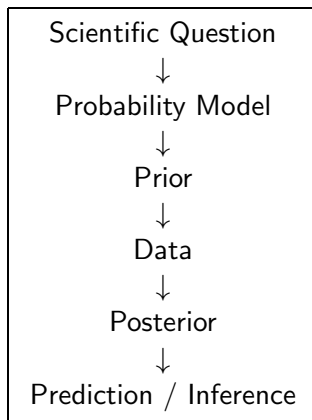
- What is the most likely outcome?
- How uncertain is the prediction?
- What happens when new information arrives?

Bayesian methods naturally represent

prediction + uncertainty.

The Bayesian Workflow

A typical Bayesian analysis looks like



We will build each part during this course.

What Makes Bayesian Statistics Different?

A Bayesian analysis allows us to talk directly about uncertainty in the unknown quantity.

For example:

$$P(\theta > 0 \mid y) = 0.95.$$

This can be interpreted as

Given the model and data, there is a 95% posterior probability that $\theta > 0$.

This is one of the central interpretations we will study.

From Bayesian Thinking to Probability

So far, we have used ideas such as

$$P(A)$$

and

$$P(A \mid B).$$

But what exactly is a probability?

What is an event?

How do we define conditional probability formally?

These questions lead us to probability theory.

The Mathematical Foundation

To build Bayesian models, we need to understand

$$(\Omega, \mathcal{F}, P)$$

where

- Ω is the sample space,
- $\mathcal{F}$ is the collection of events,
- P is the probability measure.

This is where Lecture 2 begins.

Lecture 2: Probability Review

Next lecture we will study:

- sample spaces and events,
- probability axioms,
- conditional probability,
- independence,
- Bayes' theorem,
- simulation in R.

Bayesian Thinking $\longrightarrow$ Probability $\longrightarrow$ Bayesian Inference

Three Ideas to Remember

- ① Statistics is about learning about unknown quantities using data.
- ② Bayesian inference combines prior information and data:

$$\text{Prior} + \text{Data} \longrightarrow \text{Posterior}$$

- ③ Conditional probability and Bayes' theorem provide the mathematical foundation for Bayesian learning.

One Last Question

Suppose you observe

$H, H, H, H, H, H, H, H, H, H.$

What should you believe about

$$\theta = P(H)?$$

- Is $\theta = 0.5$ still plausible?
- Is $\theta = 1$ certain?
- How much should the data change our beliefs?

These are Bayesian questions.

After-class Reading

Peter D. Hoff

A First Course in Bayesian Statistical Methods

Read the introductory material on

- belief and probability,
- Bayes' rule,
- Bayesian reasoning.

Next lecture: Probability Review I